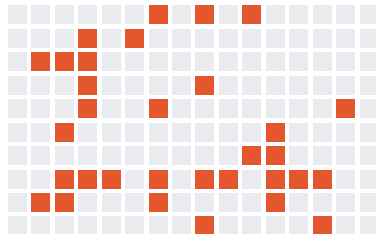


MASTER'S THESIS · BOCCONI UNIVERSITY

Imputation of Missing Data in Financial Time Series

From reconstruction error to downstream decisions



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ACADEMIC YEAR

2025/2026

Financial data is never fully observed

THE PROBLEM

Gaps are everywhere

halts, dropped quotes, illiquid tenors

The usual fixes distort risk

stale prices, lost dates, damped correlations

Evaluation is pointwise

MAE or RMSE on the masked cells only

RESEARCH QUESTIONS

1

Mechanism or amount?

2

Do deep imputers earn their cost?

3

Is pointwise accuracy a good proxy?

4

Do rankings survive downstream?

Aim. Score each imputation on reconstruction *and* on the model built on it, against an oracle fitted on the complete panel.

A controlled experiment on a complete equity panel



Daily log returns

not prices



Offline reconstruction

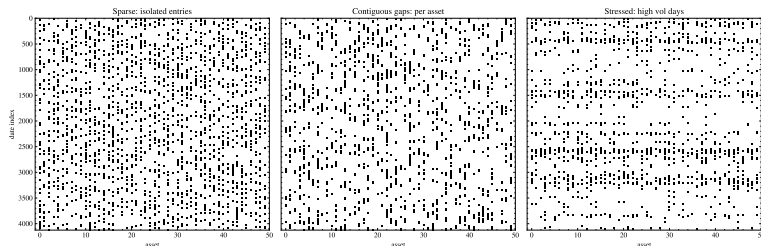
the full sample is available



Known truth

for every deleted cell

Three mechanisms, matched on the share of missing cells



Sparse MCAR

isolated cells

Contiguous gaps MCAR

runs within one asset

Stressed MNAR

the most volatile dates

Expected share of missing cells		0.1%	1%	5%	10%	30%
Sparse	r	0.001	0.01	0.05	0.1	0.3
Gaps	(n, ℓ)	(20, 10)	(200, 10)	(1000, 10)	(1000, 20)	(2000, 30)
Stressed	(r, h, p)	(0, 0.1, 0.99)	(0.005, 0.5, 0.99)	(0.04, 0.2, 0.95)	(0.05, 0.5, 0.9)	(0.2, 1, 0.9)

Seven methods, scored on the whole panel

SEVEN IMPUTERS

Baselines	Deletion, Zero
Classical	KNN, Low-rank, EM
Deep learning	BRITS (RNN), CSDI (diffusion)

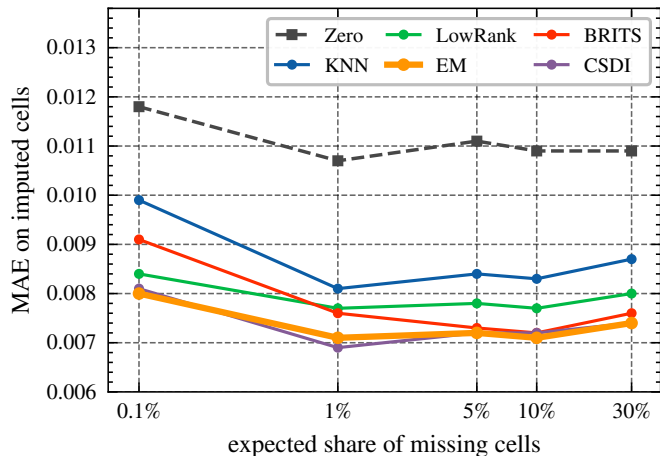
$$\mathbb{E}[r_{t,m} \mid r_{t,o}] = \mu_m + \Sigma_{mo} \Sigma_{oo}^{-1} (r_{t,o} - \mu_o)$$

EM fill: conditional mean given same-date observations

SEVEN METRICS, TWO DOWNSTREAM MODELS

Pointwise	MAE, RMSE
Marginal	VaRErr, WassErr
Second moment	CovErr
Temporal	AbsACFErr, LevErr
Downstream	Tangency portfolio Factor structure, D_k

Pointwise errors hit a floor at once, and EM sits on it



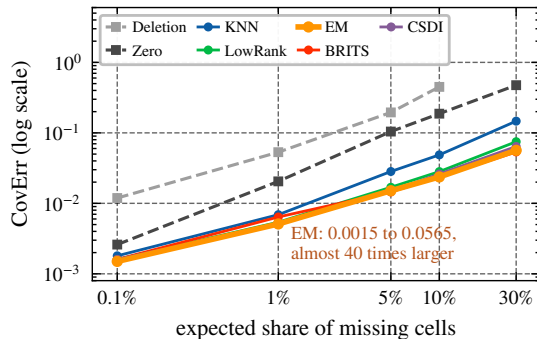
EM, MAE ACROSS SPARSE LEVELS

0.0071 to
0.0080

- Information is cross-sectional
- Conditional methods differ in the third decimal
- EM is rarely beaten

Sparse pattern, MAE on the imputed cells (Table 3.1).

Accurate cells do not make an accurate panel



almost 40×

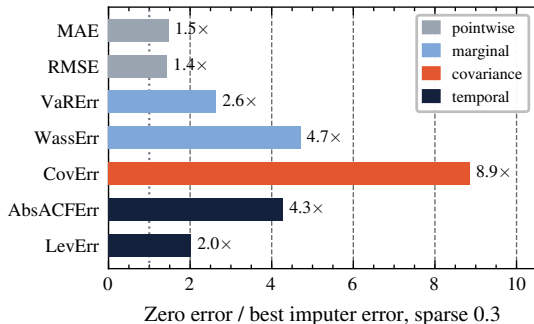
growth of EM's CovErr, MAE flat

1.5×

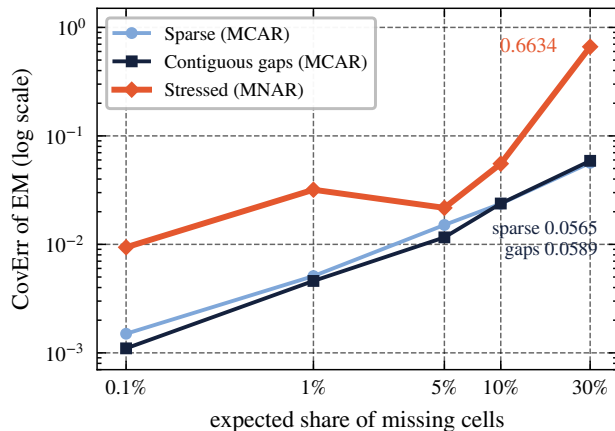
Zero's MAE vs the best, sparse 0.3

8.9×

Zero's CovErr vs the best, sparse 0.3



The mechanism matters more than the amount

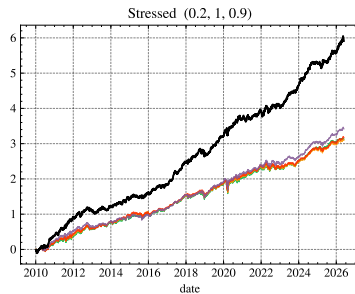
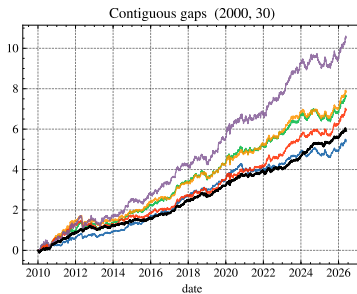
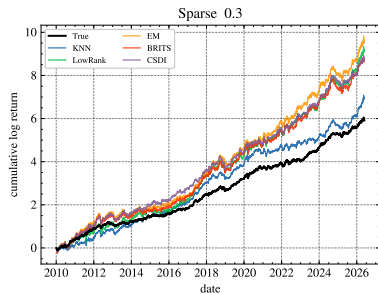


EM CovErr	Sparse	Gaps	Stressed
10% missing	0.0237	0.0238	0.0554
30% missing	0.0565	0.0589	0.6634

- MCAR patterns: interchangeable
- Stressed MNAR removes the conditioning set
- At the heaviest level no method separates

EM, covariance error by pattern at matched shares (Tables 3.1 to 3.3).

Downstream I: the portfolio amplifies the distortion



ORACLE SHARPE

1.636

HEAVY MCAR: VARIANCE

0.049 → 0.29

HEAVY MNAR: CUM. RETURN

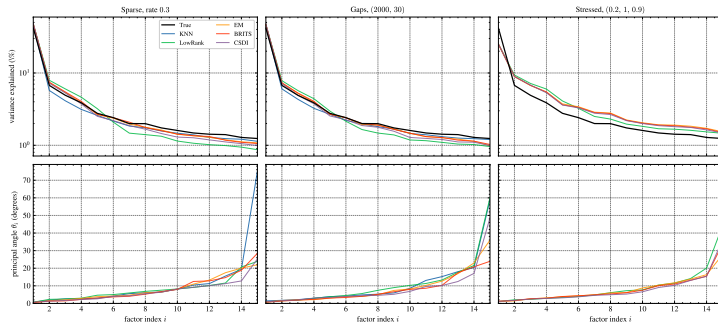
just over half

SHARPE, BRITS VS EM

1.30 vs 1.51

In-sample diagnostic of reconstruction fidelity, not investment performance.

Downstream II: a robust market factor, a fragile subspace



Variance shares (top) and principal angles (bottom), heaviest level of each pattern.

Market factor, D_1 < 0.026

Five factors, D_5 **0.14 to 0.39**

PC1 share, true 41.7%

PC1 share, imputed 45.6 to 48.5%

Low-rank worst D_5 10 of 15

$\hat{\Sigma}^{-1} = \sum_i \hat{\lambda}_i^{-1} \hat{v}_i \hat{v}_i^\top$
 portfolio weight sits on the
 worst-recovered directions

The deep models did not earn their cost

1 s

EM
per panel

1 h

BRITS
training per panel

4 h

CSDI
training per panel, plus
24 min sampling

70 h

All neural training
15 levels, four fifths
CSDI

CSDI matches EM at about **four orders of magnitude** more computation

Single Mac mini (M4, 16 GB), CPU only. Approximate times, Section 3.3.

Conclusions

1 The **mechanism**, not the amount

2 Deep models: **no gain**, far higher cost

3 Pointwise accuracy is a **poor proxy**

4 Rankings **largely survive** downstream

Score an imputation against the use it is put to.

Limitations one panel, injected missingness, Gaussian EM, in-sample portfolio, offline

Next causal imputation, multiple imputation, tests beyond the second moment

Thank you

Backup: evaluation scores under sparse missingness (Table 3.1; best value per level in bold)

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
0.001	Deletion	–	–	0.0119	0.0001	0.0001	0.0101	0.0114
	Zero	0.0118	0.0194	0.0026	0.0000	0.0000	0.0009	0.0008
	KNN	0.0099	0.0168	0.0018	0.0000	0.0000	0.0006	0.0005
	LowRank	0.0084	0.0152	0.0016	0.0000	0.0000	0.0005	0.0004
	EM	0.0080	0.0147	0.0015	0.0000	0.0000	0.0005	0.0004
	BRITS	0.0091	0.0158	0.0016	0.0000	0.0000	0.0005	0.0004
	CSDI	0.0081	0.0151	0.0016	0.0000	0.0000	0.0005	0.0004
0.01	Deletion	–	–	0.0532	0.0005	0.0004	0.0247	0.0334
	Zero	0.0107	0.0161	0.0205	0.0001	0.0001	0.0035	0.0027
	KNN	0.0081	0.0123	0.0069	0.0001	0.0000	0.0019	0.0017
	LowRank	0.0077	0.0117	0.0054	0.0000	0.0000	0.0016	0.0015
	EM	0.0071	0.0109	0.0051	0.0000	0.0000	0.0015	0.0014
	BRITS	0.0076	0.0116	0.0064	0.0001	0.0000	0.0017	0.0016
	CSDI	0.0069	0.0107	0.0050	0.0000	0.0000	0.0015	0.0014
0.05	Deletion	–	–	0.1951	0.0019	0.0012	0.1334	0.0515
	Zero	0.0111	0.0167	0.1044	0.0005	0.0006	0.0132	0.0073
	KNN	0.0084	0.0127	0.0283	0.0003	0.0002	0.0047	0.0045
	LowRank	0.0078	0.0118	0.0169	0.0002	0.0001	0.0045	0.0041
	EM	0.0072	0.0113	0.0151	0.0002	0.0002	0.0044	0.0039
	BRITS	0.0073	0.0113	0.0145	0.0002	0.0002	0.0043	0.0040
	CSDI	0.0072	0.0112	0.0149	0.0002	0.0002	0.0051	0.0041
0.1	Deletion	–	–	0.4492	0.0091	0.0051	–	–
	Zero	0.0109	0.0163	0.1872	0.0011	0.0011	0.0276	0.0111
	KNN	0.0083	0.0125	0.0487	0.0007	0.0004	0.0090	0.0067
	LowRank	0.0077	0.0116	0.0282	0.0004	0.0003	0.0072	0.0058
	EM	0.0071	0.0110	0.0237	0.0005	0.0003	0.0068	0.0056
	BRITS	0.0072	0.0111	0.0238	0.0005	0.0003	0.0068	0.0057
	CSDI	0.0072	0.0111	0.0261	0.0004	0.0003	0.0077	0.0057
0.3	Deletion	–	–	–	–	–	–	–
	Zero	0.0109	0.0161	0.4749	0.0037	0.0033	0.0638	0.0201
	KNN	0.0087	0.0129	0.1467	0.0024	0.0014	0.0173	0.0118
	LowRank	0.0080	0.0119	0.0749	0.0014	0.0007	0.0170	0.0105
	EM	0.0074	0.0112	0.0565	0.0017	0.0009	0.0150	0.0099
	BRITS	0.0076	0.0114	0.0536	0.0015	0.0008	0.0149	0.0101
	CSDI	0.0074	0.0113	0.0641	0.0014	0.0008	0.0203	0.0112

Backup: evaluation scores under contiguous-gap missingness (Table 3.2; best value per level in bold)

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
(20, 10)	Deletion	—	—	0.0103	0.0001	0.0001	0.0033	0.0033
	Zero	0.0121	0.0184	0.0027	0.0000	0.0000	0.0003	0.0002
	KNN	0.0096	0.0144	0.0016	0.0000	0.0000	0.0002	0.0001
	LowRank	0.0087	0.0123	0.0010	0.0000	0.0000	0.0002	0.0001
	EM	0.0079	0.0117	0.0011	0.0000	0.0000	0.0001	0.0001
	BRITS	0.0078	0.0112	0.0010	0.0000	0.0000	0.0001	0.0001
	CSDI	0.0079	0.0115	0.0009	0.0000	0.0000	0.0001	0.0001
(200, 10)	Deletion	—	—	0.1492	0.0010	0.0004	0.0243	0.0158
	Zero	0.0107	0.0156	0.0222	0.0001	0.0001	0.0037	0.0015
	KNN	0.0082	0.0121	0.0065	0.0001	0.0000	0.0016	0.0012
	LowRank	0.0078	0.0118	0.0061	0.0001	0.0000	0.0015	0.0012
	EM	0.0072	0.0109	0.0046	0.0001	0.0000	0.0014	0.0011
	BRITS	0.0073	0.0111	0.0049	0.0001	0.0000	0.0015	0.0011
	CSDI	0.0072	0.0109	0.0049	0.0001	0.0000	0.0015	0.0011
(1000, 10)	Deletion	—	—	0.1825	0.0017	0.0010	0.1135	0.0493
	Zero	0.0104	0.0152	0.0785	0.0004	0.0005	0.0155	0.0048
	KNN	0.0080	0.0118	0.0224	0.0003	0.0002	0.0059	0.0032
	LowRank	0.0074	0.0108	0.0137	0.0002	0.0001	0.0048	0.0029
	EM	0.0069	0.0103	0.0116	0.0002	0.0001	0.0049	0.0028
	BRITS	0.0070	0.0104	0.0116	0.0002	0.0001	0.0049	0.0028
	CSDI	0.0069	0.0103	0.0114	0.0002	0.0001	0.0051	0.0027
(1000, 20)	Deletion	—	—	0.8524	0.0087	0.0042	0.1872	0.1681
	Zero	0.0110	0.0163	0.1716	0.0010	0.0010	0.0354	0.0068
	KNN	0.0085	0.0127	0.0487	0.0007	0.0004	0.0117	0.0049
	LowRank	0.0078	0.0119	0.0310	0.0004	0.0003	0.0088	0.0046
	EM	0.0073	0.0114	0.0238	0.0005	0.0003	0.0083	0.0044
	BRITS	0.0074	0.0114	0.0241	0.0005	0.0003	0.0082	0.0045
	CSDI	0.0073	0.0115	0.0328	0.0004	0.0003	0.0095	0.0044
(2000, 30)	Deletion	—	—	1.4799	0.0104	0.0100	—	—
	Zero	0.0108	0.0161	0.4324	0.0028	0.0027	0.0908	0.0143
	KNN	0.0085	0.0128	0.1378	0.0018	0.0011	0.0277	0.0090
	LowRank	0.0080	0.0120	0.0820	0.0010	0.0006	0.0218	0.0083
	EM	0.0073	0.0113	0.0589	0.0012	0.0007	0.0212	0.0079
	BRITS	0.0074	0.0114	0.0555	0.0011	0.0006	0.0187	0.0082
	CSDI	0.0074	0.0115	0.0896	0.0010	0.0007	0.0265	0.0091

Backup: evaluation scores under stressed missingness (Table 3.3; best value per level in bold)

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
(0, 0.1, 0.99)	Deletion	–	–	0.2813	0.0009	0.0004	0.0582	0.0200
	Zero	0.0552	0.0640	0.0676	0.0001	0.0001	0.0115	0.0070
	KNN	0.0202	0.0285	0.0229	0.0000	0.0000	0.0034	0.0029
	LowRank	0.0158	0.0220	0.0116	0.0000	0.0000	0.0025	0.0020
	EM	0.0133	0.0187	0.0094	0.0000	0.0000	0.0021	0.0017
	BRITS	0.0301	0.0385	0.0253	0.0000	0.0000	0.0046	0.0036
	CSDI	0.0163	0.0226	0.0175	0.0000	0.0000	0.0027	0.0022
(0.005, 0.5, 0.99)	Deletion	–	–	0.2960	0.0010	0.0005	0.0687	0.0268
	Zero	0.0321	0.0463	0.2238	0.0005	0.0003	0.0483	0.0172
	KNN	0.0161	0.0261	0.0991	0.0001	0.0001	0.0168	0.0095
	LowRank	0.0129	0.0200	0.0279	0.0000	0.0001	0.0078	0.0067
	EM	0.0118	0.0189	0.0319	0.0000	0.0001	0.0077	0.0065
	BRITS	0.0122	0.0197	0.0401	0.0000	0.0001	0.0082	0.0069
	CSDI	0.0136	0.0233	0.0751	0.0000	0.0001	0.0132	0.0087
(0.04, 0.2, 0.95)	Deletion	–	–	0.5325	0.0043	0.0018	0.1079	0.0491
	Zero	0.0143	0.0219	0.2254	0.0010	0.0007	0.0375	0.0143
	KNN	0.0092	0.0141	0.0562	0.0003	0.0002	0.0092	0.0067
	LowRank	0.0083	0.0125	0.0253	0.0002	0.0001	0.0068	0.0052
	EM	0.0077	0.0117	0.0217	0.0002	0.0001	0.0062	0.0048
	BRITS	0.0078	0.0118	0.0231	0.0002	0.0001	0.0064	0.0049
	CSDI	0.0078	0.0122	0.0374	0.0002	0.0001	0.0078	0.0061
(0.05, 0.5, 0.9)	Deletion	–	–	0.6318	0.0056	0.0023	0.1230	0.0630
	Zero	0.0174	0.0249	0.5125	0.0030	0.0017	0.0843	0.0244
	KNN	0.0103	0.0161	0.1602	0.0010	0.0004	0.0258	0.0124
	LowRank	0.0091	0.0139	0.0483	0.0004	0.0002	0.0110	0.0091
	EM	0.0084	0.0130	0.0554	0.0008	0.0003	0.0107	0.0087
	BRITS	0.0106	0.0162	0.2083	0.0015	0.0006	0.0256	0.0126
	CSDI	0.0086	0.0137	0.0542	0.0008	0.0003	0.0165	0.0109
(0.2, 1, 0.9)	Deletion	–	–	–	–	–	–	–
	Zero	0.0149	0.0219	0.7677	0.0070	0.0042	0.1201	0.0333
	KNN	–	–	–	–	–	–	–
	LowRank	0.0137	0.0212	0.6525	0.0062	0.0029	0.1111	0.0316
	EM	0.0134	0.0210	0.6634	0.0065	0.0031	0.1112	0.0315
	BRITS	0.0136	0.0214	0.6480	0.0063	0.0025	0.1069	0.0299
	CSDI	0.0134	0.0210	0.6559	0.0064	0.0028	0.1065	0.0298

Backup: optimal portfolio at every level (Tables 3.4 to 3.6; best Sharpe in bold)

Expected share	0.1%	1%	Sparse			0.1%	Contiguous gaps				0.1%	1%	Stressed		
			5%	10%	30%		1%	5%	10%	30%			5%	10%	30%
Sharpe ratio (true 1.636)															
KNN	1.635	1.623	1.569	1.490	1.279	1.635	1.627	1.575	1.500	1.186	1.619	1.565	1.542	1.418	–
LowRank	1.635	1.624	1.565	1.529	1.253	1.635	1.627	1.593	1.560	1.225	1.629	1.606	1.586	1.494	1.195
EM	1.634	1.625	1.567	1.531	1.259	1.635	1.628	1.596	1.549	1.250	1.631	1.612	1.587	1.506	1.207
BRITS	1.634	1.625	1.572	1.542	1.226	1.635	1.629	1.588	1.555	1.276	1.620	1.602	1.593	1.304	1.212
CSDI	1.635	1.626	1.580	1.527	1.279	1.635	1.628	1.589	1.542	1.186	1.630	1.566	1.580	1.508	1.281
L^1 weight distance to the oracle															
KNN	0.30	1.11	2.92	4.07	8.43	0.16	0.87	2.24	4.31	6.52	1.13	2.50	3.16	4.64	–
LowRank	0.29	1.08	3.01	3.33	12.21	0.11	0.84	1.91	3.14	9.14	0.76	1.58	2.09	3.50	5.30
EM	0.30	1.03	3.06	3.40	13.76	0.10	0.84	1.77	3.40	10.17	0.60	1.44	2.00	3.45	5.22
BRITS	0.33	0.94	2.81	3.10	12.61	0.10	0.73	1.99	3.20	7.50	1.20	1.78	1.91	4.49	5.43
CSDI	0.30	1.02	2.89	3.03	11.33	0.11	0.79	1.97	3.52	14.48	0.66	2.91	2.62	3.50	5.22
Annualised mean (true 0.361)															
KNN	0.362	0.358	0.373	0.351	0.422	0.363	0.363	0.372	0.380	0.329	0.347	0.328	0.359	0.346	–
LowRank	0.362	0.360	0.391	0.375	0.556	0.362	0.365	0.369	0.386	0.465	0.356	0.353	0.365	0.359	0.191
EM	0.362	0.361	0.391	0.377	0.581	0.362	0.363	0.371	0.379	0.478	0.356	0.353	0.370	0.359	0.189
BRITS	0.362	0.359	0.384	0.371	0.528	0.362	0.362	0.369	0.358	0.423	0.345	0.349	0.369	0.243	0.193
CSDI	0.362	0.361	0.396	0.373	0.531	0.362	0.364	0.374	0.381	0.643	0.360	0.375	0.392	0.354	0.209
Annualised variance (true 0.0488)															
KNN	0.0491	0.0486	0.0564	0.0554	0.1086	0.0492	0.0497	0.0557	0.0644	0.0767	0.0459	0.0439	0.0543	0.0597	–
LowRank	0.0490	0.0492	0.0625	0.0603	0.1970	0.0491	0.0503	0.0537	0.0612	0.1439	0.0477	0.0483	0.0529	0.0576	0.0255
EM	0.0491	0.0493	0.0623	0.0607	0.2130	0.0491	0.0499	0.0539	0.0598	0.1460	0.0478	0.0480	0.0545	0.0568	0.0245
BRITS	0.0492	0.0487	0.0597	0.0577	0.1852	0.0490	0.0494	0.0540	0.0529	0.1098	0.0452	0.0476	0.0536	0.0347	0.0255
CSDI	0.0489	0.0492	0.0627	0.0597	0.1723	0.0489	0.0499	0.0555	0.0611	0.2937	0.0487	0.0572	0.0615	0.0551	0.0266

Backup: factor structure at every level (Tables 3.7 to 3.9; best D_5 in bold)

Expected share	0.1%	1%	Sparse			0.1%	Contiguous gaps				0.1%	1%	Stressed		
			5%	10%	30%		1%	5%	10%	30%			5%	10%	30%
Variance share of the first component (%) (true 41.74)															
KNN	41.76	41.87	42.28	43.00	45.62	41.75	41.90	42.29	42.93	44.63	41.31	39.81	41.68	40.27	–
LowRank	41.77	41.91	42.64	43.60	47.56	41.75	41.93	42.42	43.63	46.84	41.75	41.92	42.73	43.50	25.12
EM	41.77	41.91	42.64	43.59	47.88	41.75	41.92	42.42	43.52	46.95	41.73	41.80	42.66	43.29	24.81
BRITS	41.77	41.87	42.59	43.47	47.09	41.75	41.92	42.37	43.44	46.36	41.45	41.56	42.58	40.36	25.19
CSDI	41.77	41.91	42.72	43.73	48.54	41.75	41.92	42.43	43.78	47.94	42.06	43.62	43.27	44.63	25.01
Variance share of the first five (%) (true 60.02)															
KNN	60.04	60.09	60.32	60.35	61.19	60.02	60.15	60.21	60.62	60.92	59.70	58.28	59.49	57.71	–
LowRank	60.06	60.26	61.37	62.56	69.35	60.03	60.26	61.07	62.50	67.76	60.12	60.67	61.36	63.34	51.75
EM	60.06	60.22	61.19	62.05	67.16	60.03	60.24	60.88	62.14	66.09	60.07	60.14	61.03	61.77	49.58
BRITS	60.05	60.16	61.14	61.96	66.74	60.03	60.24	60.84	62.08	65.70	59.73	59.92	60.96	59.06	50.00
CSDI	60.06	60.22	61.18	62.03	67.28	60.03	60.24	60.85	62.22	66.31	60.23	61.24	61.34	62.49	49.66
Subspace distance D_1															
KNN	0.0006	0.0030	0.0082	0.0133	0.0252	0.0008	0.0034	0.0089	0.0179	0.0511	0.0106	0.0309	0.0169	0.0398	–
LowRank	0.0006	0.0026	0.0068	0.0100	0.0215	0.0004	0.0031	0.0053	0.0143	0.0375	0.0079	0.0180	0.0149	0.0268	0.1186
EM	0.0005	0.0026	0.0062	0.0094	0.0152	0.0005	0.0021	0.0047	0.0097	0.0262	0.0063	0.0178	0.0126	0.0232	0.1148
BRITS	0.0005	0.0034	0.0054	0.0100	0.0161	0.0004	0.0024	0.0046	0.0104	0.0252	0.0176	0.0203	0.0132	0.1407	0.1106
CSDI	0.0006	0.0027	0.0062	0.0122	0.0211	0.0003	0.0028	0.0040	0.0158	0.0465	0.0088	0.0316	0.0170	0.0303	0.1141
Subspace distance D_5															
KNN	0.0051	0.0143	0.0364	0.0436	0.1434	0.0040	0.0117	0.0376	0.0525	0.1365	0.0276	0.1055	0.0680	0.1131	–
LowRank	0.0036	0.0195	0.0809	0.2551	0.3914	0.0033	0.0207	0.0334	0.1892	0.3291	0.0152	0.0555	0.0714	0.3293	0.3814
EM	0.0040	0.0126	0.0440	0.1120	0.2039	0.0034	0.0117	0.0249	0.0550	0.1273	0.0145	0.0950	0.0327	0.0843	0.1343
BRITS	0.0046	0.0154	0.0347	0.1034	0.2376	0.0029	0.0115	0.0270	0.0508	0.1177	0.0317	0.0900	0.0368	0.1219	0.1184
CSDI	0.0041	0.0126	0.0323	0.1014	0.1954	0.0027	0.0100	0.0291	0.0453	0.1435	0.0160	0.1166	0.0383	0.0964	0.1340